

Nikola Staykov

Robotics, Systems and Control Student @ ETH Zurich

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Education

ETH Zurich

M.Sc. in Robotics, Systems, and Control Zurich, Switzerland 9/2024 - 2/2027
Key Courses: Robot Dynamics; Probabilistic AI; Machine Perception

Technical University of Munich

B.Sc. in Engineering Science Munich, Germany 10/2021 - 9/2024
Key Courses: Control Theory; Modelling and Simulation; Introduction to AI

Experience

Sevensense at ABB

Robotics Test Engineering Intern Zurich, Switzerland 7/2025 - 3/2026

- Worked on an in-house test automation framework, improving test coverage and minimizing manual testing effort.
- Integrated the testing framework with in a CI/CD workflow.
- Streamlined the deployment of our SLAM system on various wheeled testing platforms.
- Led the release testing efforts for a client.

Data Machine Intelligence

Simulation Engineer Munich, Germany 5/2023 - 9/2024

- Developed a flight simulation pipeline with a parallelized architecture, which enabled machine learning applications.
- Trained a reinforcement learning PPO agent for autonomous flight.
- Demonstrated the capabilities of our custom flight simulation pipeline in front of investors and clients.

Projects

Semantically Safe Robot Manipulation: From Semantic Scene Understanding to Motion Safeguards

SLAM / Barrier Functions / Python / ROS

Project Site, Paper

- Deployed an object detection and 3D object segmentation pipeline.
- Worked on a safety filter framework to certify robot inputs with respect to semantically defined constraints.
- Integrated semantic constraints in the control flow.

Note: Paper accepted to RA-L

Efficient Environment Representation for Robot Control

Sensor Fusion / SLAM / Camera Calibration / ROS / C++

- Performed sensor calibration for RGB, depth and IMU sensors.
- Utilized SLAM and sensor fusion techniques. Created an evaluation pipeline, integrating a MoCap system as a baseline.
- Generated semantically rich 3D point clouds. Enabled use in LLM safety applications.

Meinung (TUM JA Scholarship 2023/24)

NLP / Sentiment Analysis / Python

Project Site

- Moral stances analyzed in textual data from various news outlets.
- Extended the eMFD moral categories approach. Added context-related classification of texts using foundational models.
- Improved the interpretability and usability of the eMFD approach.

Awards and Recognitions

ETH Robotics Summer School; TUMJA Scholarship; Deutschlandstipendium'22/23/24; RSI'20